SATVIK PRASAD

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EDUCATION

Georgia Institute of Technology — Atlanta, GA
B.S. in Computer Engineering, Minor in Mathematics — *Aug. 2025 – May 2028*

- John Martinson Honors Program, GT Venture Capital Club.
- 'Terra Trends' Data Visualisation Team Lead, *Big Data Big Impacts@GT*

Normanhurst Boys High School — Sydney, NSW, Aus
Valedictorian, Highest Graduating Rank in Australia — *Feb. 2019 – Oct. 2024*

EXPERIENCE

Synopsis Education — May 2025 – Present
Director + CTO — *Sydney, Aus (Remote)*

- Engineered front-to-back tech stack for student-portal, landing website, and CRM + CI/CD automations.
- Developed full-stack web application in Typescript, React, PostgreSQL + Supabase, Docker, Express.js & Redis with REST web services and AWS S3.
- Contributed to 34k+ Git-tracked lines of production code.
- Wrote custom model context protocol in Python + FastMCP for quick powerpoint ideation.

Virtusa Polaris — Aug 2022
Software Engineering Intern — *Sydney, Aus*

- Learned industry-standard SCRUM techniques in the software services sector.
- Designed full-stack internal communications platform in Vue.js, GraphQL and PostgreSQL + containerized using Docker.

PROJECTS

LizaLang | *Interpreted, Dynamically-Typed Language written in OCaml* — Nov 2025 - Current

- Designing a dynamically-typed, high-performance scripting language with recursive descent parsing and semantic validation in OCaml.

I-IMO | *AI-Powered Conversational Agent — Winner @ Calhacks 12.0* — Sep 2025

- Trained a custom yolov8 model for consistent face detection and thumbnail generation with 93% face-repitition accuracy.
- Used inference hosted on Groq for chunk-based text to speech streaming, synced with visual inference.
- Vectorised transcriptions in ChromaDB for high-speed semantic searches.
- Implemented profile summaries, task generation, future meeting assistance, and prompt-based context searches using RAG.

Promptly | *World's 1st Prompt-Based Agent with System-Wide Control — Ergo Challenge @ HackGT25* — Sep 2025

- Used Omni-Parser pre-trained weights and RICO dataset to train a lightweight, local YOLO11n object detection model with 110x faster inference times.
- Utilised OpenCV for 50ms zero-shot text detection + pyautogui for system-wide control.
- Designed a feedback-driven LLM event loop for automatic task correction.
- Speech-to-text inference using PortAudio and gpt-4o-transcribe with ∼2ms inference times, outperforms Gemini 2.5 Computer Use Model by 10.5x (based on inference times).

Peggiator | *Web-based, High Performance & Extensible Music Visualiser* — Jan 2025 – May 2025

- Used a self-derived WASM implementation of the Fast-Fourier Transform & Web Workers to decompose input audio, 4x faster than V6 implementation.
- Captured system audio using a helper-binary written in Swift and ScreenCaptureKit.
- Injected audio data into a Vite + React sidebar component for the DOM.
- Used Zig and WebGL for 3D rendering of manipulated audio waveforms.
- Shipped a production-ready Desktop-version using Electron.

AWARDS

- Winner, Calhacks 12.0 @ UC Berkeley (MLH Sponsor Track – Best Use of DigitalOcean Gradient), 2000+ competitors
- Top 0.05% (Top 50) graduating rank in Australia (Achieved Maximum ATAR of 99.95 in HSC Board Examinations)
- State Rank, 10th in New South Wales for Mathematics Extension 2, High Distinction in Australian Physics & Chemistry Olympiads, Silver Medal in AIO
- Awarded Scientia Scholar and Best Mathematics Student in NSW by the University of New South Wales

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), Typescript, HTML/CSS/LaTeX, R, Golang, Zig, Rust, WebAssembly, OCaml
Frameworks: React, Node.js, Express.js, Next.js, FastMCP
Developer Tools: JIRA, Gradle, Git, Docker, NeoVim, AWS + any IDE you want me to learn
Libraries: pandas, NumPy, Matplotlib, Raylib, OpenGL, WebGL, torch, TensorFlow, transformers, Flask